

What is space?

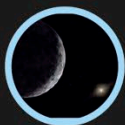
Space holds many secrets. It contains places where human beings can be stretched into different shapes, and their body fluids can be boiled, or frozen solid. That's why astronauts wear protective clothing in space. Welcome to the mysterious – and endlessly fascinating – Universe.

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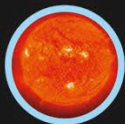
When people think of space, they think of the following:



Astronauts feel **weightless** and float around.



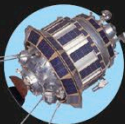
Vast areas of space are **completely empty**.



Every **star** is a burning ball of gas. Our Sun is a star.



Astronauts, or **cosmonauts**, are people who travel into space.



Space probes and **artificial satellites** are what scientists use to explore space.



There is no air in space, so there is **absolute silence**.

A planetary nebula is a colourful cloud of gas and dust ejected by a dying star. This is the Helix Nebula, about 650 light years away, seen from NASA's Spitzer Space Telescope.



Is that space?

On a cloudless night, you can see thousands of stars. Space is the name we give to the huge empty areas in between the atmospheres of stars and planets. Apart from the odd rock, space is sprinkled only with dust and gas.

Too big to imagine

Astronomers measure distance in space in light years. One light year is the distance light travels in one year – that's about 10 million million km (6 million million miles).